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Effects of Roughness on a Turbulent Boundary Layer in Hypersonic Flow

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Particle Image Velocimetry experiments were performed to study the effects of surface roughness on a hypersonic ($M = 7.2$), low Reynolds number (typically $Re_\theta = 3600$), turbulent boundary layer. Two roughness types were used: diamond mesh and square bars, with roughness heights of about 136 and 102 viscous units, respectively. The streamwise velocity fluctuation normalized by the wall friction velocity indicated that there may exist significant compressibility effects at these Mach numbers.

Nomenclature

B	additive constant on the log law
p	pressure, Pa
k	roughness element height, mm
M	Mach number
Re_θ	Reynolds number based on momentum thickness
u	fluctuating component of velocity, streamwise direction, m/s
u_τ	friction velocity, m/s
U	streamwise velocity, m/s
T	temperature, K
v	fluctuating component of velocity, wall-normal direction, m/s
x	streamwise distance, mm
y	normal distance from the wall, mm
δ	boundary layer thickness, mm
ϵ	offset in origin of the velocity profiles, mm
κ	von Karman constant
ν	kinematic viscosity, Pa $\cdot$ s
Π	Coles' wake factor
ρ	density, kg/m^3

Superscript

$+$	Indicates inner scaling

Subscript

e	Indicates value in the freestream
w	Indicates value at wall
0	Indicates stagnation value

$<>$ indicates time average

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I. Introduction

The present study is part of a broad effort to investigate the behavior of hypersonic turbulent boundary layers, including the effects of roughness and transpiration. Here, we report preliminary measurements on the effects of roughness.

For the flow over a smooth wall with a zero pressure gradient, previous experiments on hypersonic boundary layers have documented the general features of the mean flow, but even for this relatively simple flow case there are only a few studies where extensive turbulence measurements have been made. One particularly important experiment in hypersonic turbulent flows was that performed at NASA Ames.[15,19–21] The flow developed on an axisymmetric body, so that it was free of side wall effects, although there may be some residual history effects from the development over the ogive nose body. The upstream boundary layer was representative of a zero pressure gradient Mach 6.7 flow.[21] The mean velocity followed the standard semi-logarithmic profile when transformed according to Van Driest, which takes into account the density changes across the boundary layer. A similar profile was measured by Baumgartner[2] at a Mach number of 7.2, and by McGinley *et al.*[14] at a Mach number of 11. Together these data support the notion that the scaling laws for the mean flow are essentially independent of Mach number.

The scaling with respect to turbulence behavior is not so clear. In low-speed flows, the turbulence scales much like the mean flow, in that the near-wall region scales with the friction velocity $u_\tau = \sqrt{\tau_w/\rho}$ and the viscous length scale ν/u_τ (although the peak value of the streamwise turbulence intensity shows a slow increase with Reynolds number), whereas the outer region scales with u_τ and the boundary layer thickness δ. It was proposed by Morkovin[16] that in high-speed flows, turbulence ought to scale with the density weighted velocity scale $u_\tau\sqrt{\rho_w/\rho}$ instead of u_τ, so that the scaling is applied to the turbulent stresses rather than the intensities. For the outer region, we still have the outer length scale, δ, and for the inner region, the length scale is now based on the kinematic viscosity at the wall, that is, ν_w/u_τ. Morkovin showed that this scaling collapsed the streamwise turbulence intensity for Mach numbers up to 4.5(see, for example Smits & Dussauge[28]).

Owen *et al.*[21] and McGinley *et al.*[14] obtained turbulence measurements using hot wire anemometry, and they are two of the very few data sets on turbulence in hypersonic flows. McGinley *et al.*[14] reviewed the previous attempts to measure turbulent fluctuations in such flows, and made the observation that much of this work suffered from poor frequency response and/or suspect calibrations. It was clear from their data, however, that the turbulence intensities did not scale according to Morkovin. It is not obvious whether this was due to the difficulties in using hot-wire anemometry at high Mach numbers, or through the use of the Strong Reynolds Analogy (SRA)[28] in transforming hot wires measured mass flux intensities to velocity fluctuations, or whether it was revealing new flow physics associated with high Mach number turbulence.

As to the effects of roughness on high-speed boundary layers, only two relatively complete experiments have been reported. The study by Berg (1977) includes mean and fluctuating measurements for sudden changes in surface roughness, from smooth to rough, and from rough to smooth. The freestream Mach number was approximately 6, and the wall was near adiabatic. The roughness was formed by square bars, with a pitch to height ratio $\lambda/k = 4$. The roughness heights varied from 0.0125, 0.025 and 0.05 in (0.32, 0.64, and 1.27 mm), corresponding to $k/\delta = 0.012$, 0.025, and 0.04 (and $k^+ = ku_\tau/\nu_w = 7.1$, 14.9, and and 33.8) in the "equilibrium" regime far downstream of the step change in roughness. The effective origin distance of the profiles below the crest of the roughness element was found to be about $0.5k$. The case of $k^+ = 33.8$ was considered to be fully rough, and here the equivalent sand grain roughness k_s was found to be $1.3k$, which was about half the value found in subsonic investigations of similar roughness.

Ekoto *et al.*[5] studied the effects of large-scale ($k_s^+ \approx 100$, $k/\delta \approx 0.067$), three-dimensional, periodic surface roughness on a Mach 2.86, high Reynolds number ($Re_\theta \approx 60,000$), turbulent boundary layer on an adiabatic wall. Diamond mesh and square block ($\lambda/k = 4$) roughness geometries were used. For the mesh roughness, the flow was aligned with the long dimension of the diamond. The mesh roughness produced a pattern of oblique shocks and expansion waves leading to strong local distortions which had a significant effect on the mean and turbulent flow structure throughout the boundary layer. By comparison, the effects of the local distortions caused by the square block roughness was much less significant. The results showed that if roughness-induced localized distortions are weak, the classic inner-scaling captures the effects of the roughness on the turbulence, without the use of Morkovin scaling. The equivalent sand grain roughness k_s for the square block geometry was found to be $0.73k$, but the diamond mesh results were strongly affected by the local disturbances produced by the mesh and only tentative conclusions could be made regarding k_s.

To further investigate the effects of roughness in hypersonic turbulent boundary layers, we are performing

experiments at Mach 7.2 on a flat plate model with two different types of roughness: a diamond mesh roughness, and a two-dimensional square bar roughness with $\lambda/k = 5$. For these cases, $k^+ = 136$ and 102, respectively. PIV is used to measure the mean and turbulent velocities. The Reynolds number is relatively low ($3260 < Re_\theta < 4450$) so that the flow is amenable to DNS.

II. Experimental Facilities

The experiments are conducted on a flat plate model mounted in the Mach 8 Hypersonic Boundary Layer Facility at the Princeton Gas Dynamics Laboratory. Air is used as the primary working fluid. The maximum stagnation pressure is 10.3×10^6 Pa (1500 psi), and the maximum stagnation temperature is 870K (1100F), and the tunnel can be run up to 120 s continuously. The tunnel operating conditions give a Reynolds number range so that, at the lower Reynolds number the flow is laminar (even on the nozzle walls), and at the highest Reynolds number fully turbulent boundary layers are generated on the flat plate. A schematic of the tunnel is shown in figure 1, and it is described in more detail by Baumgartner.[2]

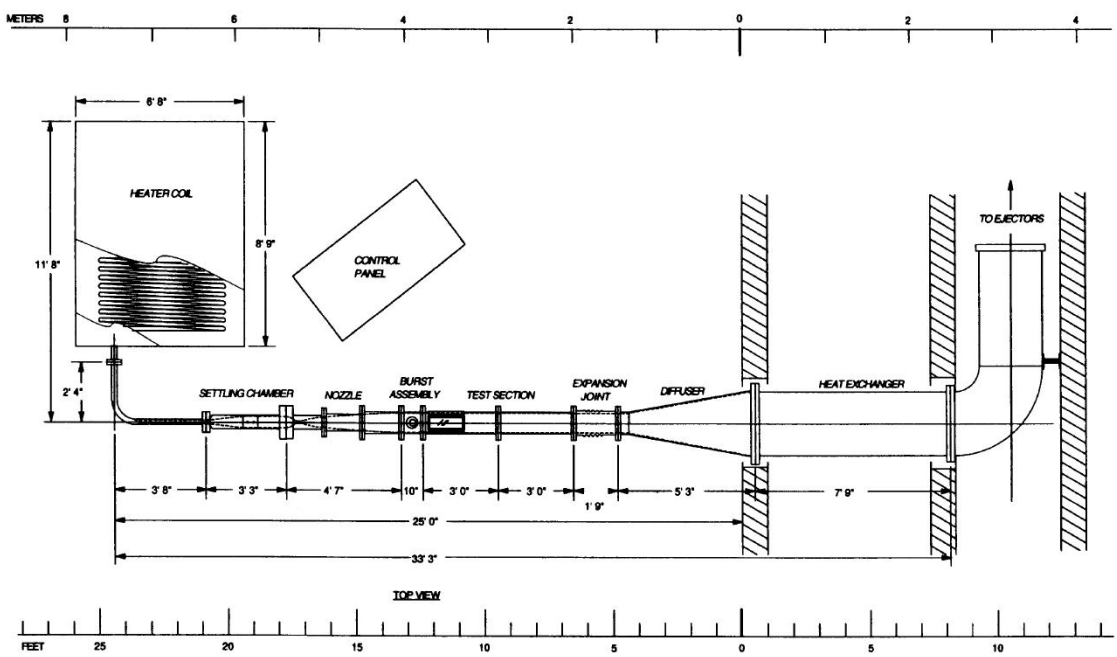

Figure 1. Schematic of the Mach 8 wind tunnel.

The test section is made up of two 914 mm (3 ft) long, 229 mm (9 in) inside diameter stainless steel sections. One section is fitted with four orthogonal window cavities. The cavities are 127 mm (5 in) x 206 mm (16 in) rectangular sections, beginning 89 mm (3.5 in) from the beginning of the section. The windows are recessed 38 mm (1.5 in) from the wall of the test section. The flat plate model is mounted on the centerline of the test section on a support that is fastened to a solid stainless steel window plate that bolts to the bottom window cavity. The support has a diamond-shaped cross-section to minimize flow blockage. The top and side window cavities were used for optical access to the test section. The windows are 225 mm x 137 mm x 12.7 mm in dimension and made of quartz. They are mounted into stainless steel window plates that fit over the window cavity. In previous experiments in this facility,[4] the window cavities were found to cause disturbances that were detrimental to the starting of the tunnel. Ramps were installed at the end of the window plates to help alleviate this problem, and a similar approach was taken here.

The flow in the test section has been characterized by Baumgartner,[2] Magruder,[13] and Etz.[4] When the working section is empty, the Mach number is 8.0 ± 0.1 over the central 80% of the cross-sectional area. The freestream Mach number when the flat plate model is in the tunnel at the location of the measurement is 7.2.

The flat plate model, shown in Figure 2, is described in detail by Etz.[4] It is made of brass, and it is 152 mm wide, 476 mm long, and 12 mm thick. To reduce laser reflections the model was painted black. The

surface roughness of the painted model has been estimated by Baumgartner[2] to be $\leq 2\mu$m ($ku_\tau/\nu_w \approx 0.10$), so that the plate in the absence of the mesh and bar roughnesses is assumed to be smooth.

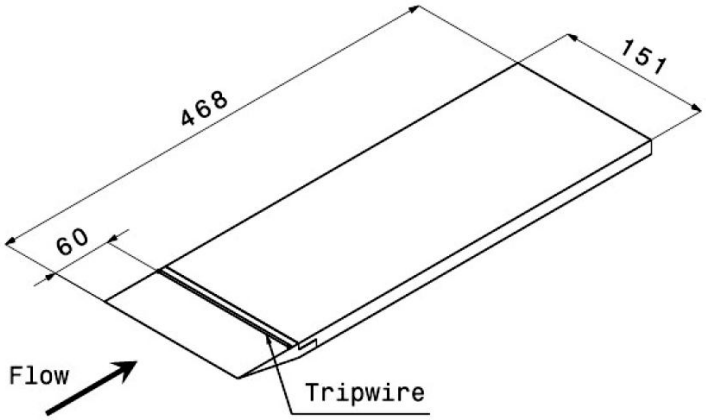

Figure 2. Schematic drawing of the flat plate model. Dimensions are in mm.

The region for the PIV measurements started approximately 380mm from the leading edge and and had a length of about 22 mm. To ensure a fully developed turbulent boundary layer at this measurement position, the flow was tripped using a tripwire 60 mm from the leading edge. The height of the tripwire was about 2.4 mm (see Baumgartner[2] for further details).

Surface temperature (T_w) and wall static pressure (p_w) measurements were made on the centerline of the model, approximately 120 mm and 380 mm downstream of the leading edge. For temperature measurements, a K-type thermocouple was used, the pressure determination was made with an Omegadyne PX309-005A5V 0-0.34 bar transducer. The settling chamber stagnation temperature (T0) and the settling chamber stagnation pressure (p_0) were measured with a K-type thermocouple and an Omega Engineering 0-137.9 bar transducer. For all thermocouples, calibrated digital panel meters (DPM) with an output of 1mV/C were used. Pressure and temperature data were digitized via a NI USB-6212 data acquisition device for further processing and logging with a LabVIEW program.

The first type of roughness was a titanium diamond mesh as shown in Figure 3. The thickness of the mesh was $k = 1.65$ mm, corresponding to $k^+ = ku_\tau/\nu_w \approx 136$, $k/\delta \approx 0.16$. The flow was aligned with the short dimension of the diamond. For the second type of roughness, a rectangular plate was machined to obtain square bar roughness (d-type roughness according to Perry *et al.*[18]). See Figure 3. The height of the square elements was the same as the mesh height (1.65 mm), with $k^+ \approx 102$, $k/\delta \approx 0.14$, and $\lambda/k = 5$.

III. Particle Image Velocimetry

The data reported here were obtained using Particle Image Velocimetry (PIV). A New Wave Tempest and Gemini PIV dual head ND:YAG laser system was used as the laser source. Each laser delivered 100 mJ energy per pulse at a wavelength of 532 nm. The laser pulses have a pulse width of 3–5 ns and a jitter of ± 0.5 ns. The time delay between the two lasers was set to 0.4 μs as suggested by Sahoo *et al.*[24]

A PCO Cooke camera with a 100 mm lens was used to acquire PIV images. Camware V2.1 was used to process the images and standard digital PIV software was used to analyze the data. The imaged region of interest was typically 16 x 16 mm.

Earlier efforts by Allen (unpublished) had shown that TiO_2 particles perform well in terms of size and reflected light intensity. Humble[10] reports that these particles have a nominal diameter of about 50 nm, but because of agglomeration the effective particle diameter is typically 400 nm, with a relaxation time of about 2.1 μs, corresponding to a frequency response of about 500 KHz.

To minimize the flow disturbance, the seeding particles were introduced in the settling chamber upstream of the throat. In the present system, the TiO_2 particles were first suspended in a fluidized bed where high pressure air (typically 10 bar above the tunnel stagnation pressure) was fed from below. A heating tape was used to heat the bottom part of the bed to help reduce particle agglomeration. The air entrained the particles, which then passed to a cyclonic separator before being injected into the stagnation chamber of the

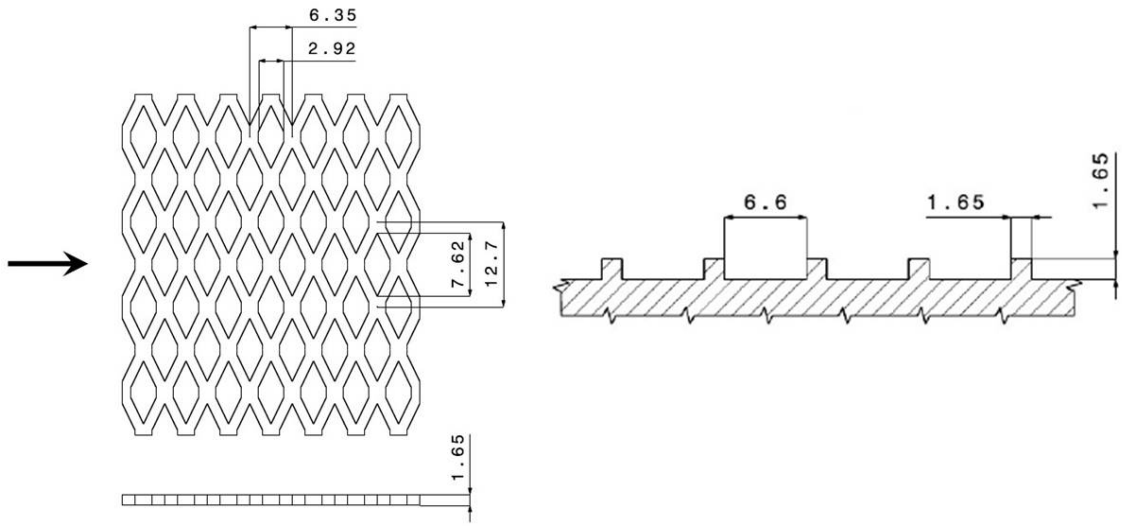

Figure 3. Left: Three-dimensional diamond mesh type roughness geometry. Right: Two-dimensional d-type square bar roughness geometry. All dimensions are in mm.

tunnel through a 12.7 mm (0.5 in) tube facing downstream. The end of the tube was approximately 450 mm upstream of the nozzle throat and was located on the centerline of the settling chamber. Further details are given by Sahoo *et al.*[24]

Seeding the flow presents significant challenges in performing accurate PIV experiments in a hypersonic flow. At hypersonic Mach numbers the air density is very low while the velocities ar every high, and both effects compound the seeding problem. The air density is a particularly important parameter, so that seeding the region close to the wall is particularly difficult.

In order to obtain accurate measurements, a number of steps were necessary in the analysis of the PIV data. First, before each run of the wind tunnel, a set of calibration images was recorded using a calibration card with a printed grid placed in the laser sheet. Second, the raw PIV images with unsatisfactory seeding density were discarded. Third, a shift and rotation was applied to the images to allow for the movement of the tunnel at start up and the misalignment of the camera (always less than 0.5°). The error when finding the correct position of the wall in each image is approximately 2 pixels which corresponds to about 0.036 mm or 0.4% of the boundary layer thickness. Fourth, a 3-step adaptive correlation calculation using successive interrogation window sizes of 128x128, 64x64, and 32x32 with 50% overlap was used to determine the instantaneous velocity vectors[26] (reducing the interrogation window further caused the results to diverge sharply). Fifth, a consistency filter that searches for one correlation peak around another within a radius of one unit was used to help eliminate bad vectors, where a unit denotes 50% of the interrogation window. The minimum number of particles was set to three. Sixth, a velocity range validation filter was applied which eliminates all vectors exceeding a certain range. Based on the flow conditions and the field of view, only vectors with an x-component between 0 px (0 m/s) and 30 px (approximately 1350 m/s) and a y-component between -10 px and 10 px (approximately 450 m/s) were accepted as valid. Finally, vectors closer than approximately 0.3 mm to the edge of the field of view were discarded because the cross-correlation does not give meaningful results when particles are entering or leaving the field of view and are only visible in one of the two images.

To calculate converged statistical quantities of the flow, an adequate data base is required. Here, the averaged flow quantities were computed using 901, 1258, and 957 pairs of images for the smooth wall, diamond mesh roughness, and square bar roughness experiments, respectively. The results were averaged over the streamwise extent of the field of view. Because of limitations on the amount of available air, the memory of the CCD camera, or the performance of the seeder, it was necessary to record the required number of PIV images during several runs spread over multiple days. To combine data from different runs, a weighted mean was used:

$$\bar{x} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i} \tag{1}$$

where $\bar{x}$ represents the weighted mean of a quantity x (e.g. the streamwise velocity), n the number of runs

and w the number of images gained from the respective run i. The velocities were first non-dimensionalized with respect to the average freestream velocity before the weighted mean was calculated.

IV. Mean Flow Results

The mean test conditions for the experiments reported here are summarized in Table 1. The mean flow conditions correspond closely to those used by Baumgartner, Magruder, Etz, and Sahoo *et al.*[2,4,13,24] As seen in Figure 4, the freestream velocity decreases slightly with distance from the wall, and therefore the boundary layer thickness was estimated by finding the point where the velocity deviates by 1% from a straight-line fit to the freestream data.[4] For the case shown in Figure 4 the boundary layer thickness is approximately 8.2 mm.

Model	p_0 (psia)	T_0 (K)	T_w (K)	p_s (psia)	M_e
smooth	$1035 \pm 1.8\%$	$756 \pm 1.5\%$	$352 \pm 2.0\%$	$0.19 \pm 1.8\%$	7.3 ± 0.1
diamond	$1026 \pm 1.9\%$	$747 \pm 1.6\%$	$338 \pm 1.7\%$	$0.16 \pm 1.8\%$	7.3 ± 0.1
square	$1030 \pm 1.9\%$	$760 \pm 1.6\%$	$337 \pm 1.6\%$	$0.17 \pm 1.6\%$	7.3 ± 0.1

Table 1. Mean test conditions. The standard deviations reflect the repeatability of the experimental conditions.

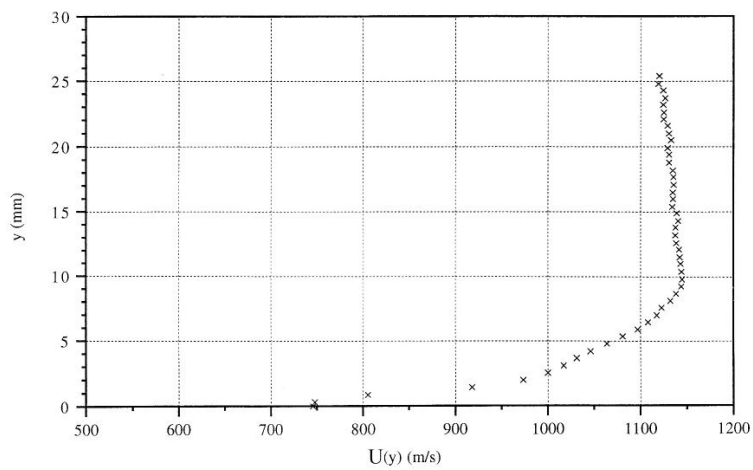

Figure 4. Mean velocity distribution in the hypersonic turbulent boundary layer, as measured by Etz[4] for $Re_\theta = 3600$.

The mean velocity profile on the smooth plate obtained by PIV was in good agreement with the Pitot probe data reported by Owen *et al.*[21] and Etz[4] (see Figure 5), although there was a slight (so far unexplained) deviation from Etz's data (obtained on the same model) between $y/\delta > 0.12$ and $y/\delta < 0.4$. The data were transformed using the Van Driest transformation,[31] assuming Walz's form of the temperature profile,[28] and the friction velocity was determined by the Clauser chart method. Here we assumed the log-law applied for $y^+ > 30$ and $y/\delta < 0.15$, where

$$\frac{U}{u_\tau} = \frac{1}{\kappa} \ln y^+ + B \tag{2}$$

where $\kappa = 0.4$ is von Kármán's constant $B = 5.1$.[a] The data for smooth plate was in good agreement with the incompressible law-of-the-wall correlation, as shown in Figure 5 and the measurements by Owen *et al.* at approximately the same Mach number.

For a rough wall, the origin of the boundary layer will not necessarily coincide with either the top of the roughness elements, or the bottom. The (positive) offset in the virtual origin below the top of the roughness elements is ϵ. In the present investigation, neither ϵ nor u_τ are known a priori, and must be found by

[a]The limits of the log-law in turbulent boundary layers is a subject of current debate, as are the values of κ and B. Neither discussion will affect the conclusions drawn here, at least to within the accuracy of the data.

$Model$	U_e	u_τ	$C_f \times 10^3$	ϵ	ν_w/u_τ	δ	θ	Re_θ	$Re_{\delta 2}$	$2\Pi/\kappa$
smooth	1094	55.8	0.219	-0.12	0.0297	7.21	0.19	3257	766	2.88
diamond	1129	153	1.26	0.55	0.0123	10.4	0.27	4450	967	1.2
square	1117	132	1.05	1.5	0.0165	12.1	0.25	3961	890	1.3

Table 2. Boundary layer parameters. All dimensions are in mm or m/s.

iteration. Here we followed the procedure first outlined by Perry and Joubert.[17] Hence, for a rough wall, including the Coles wake function

$$\frac{U}{u_\tau} = \frac{1}{\kappa}\ln y^+ + B - \Delta\left(U/u_\tau\right) + \frac{2\Pi}{\kappa}\sin^2\left(\frac{\pi}{2}\frac{y}{\delta}\right) \tag{3}$$

where $y = y_T + \epsilon$, y_T is the wall-normal distance measured from the tops of the roughness elements, and $\Delta(U/u_\tau)$ is Hama's roughness function (the amount the velocity profile on a rough wall is shifted below the standard log-law by the effects of roughness). When this procedure was applied to the smooth wall profile (where we would expect $\epsilon = 0$), a small offset of $\epsilon = -0.12$ mm was found, attributable to the laser reflections off the wall surface affecting the estimate of the wall position. The boundary layer parameters for the smooth wall case are summarized in Table 2.

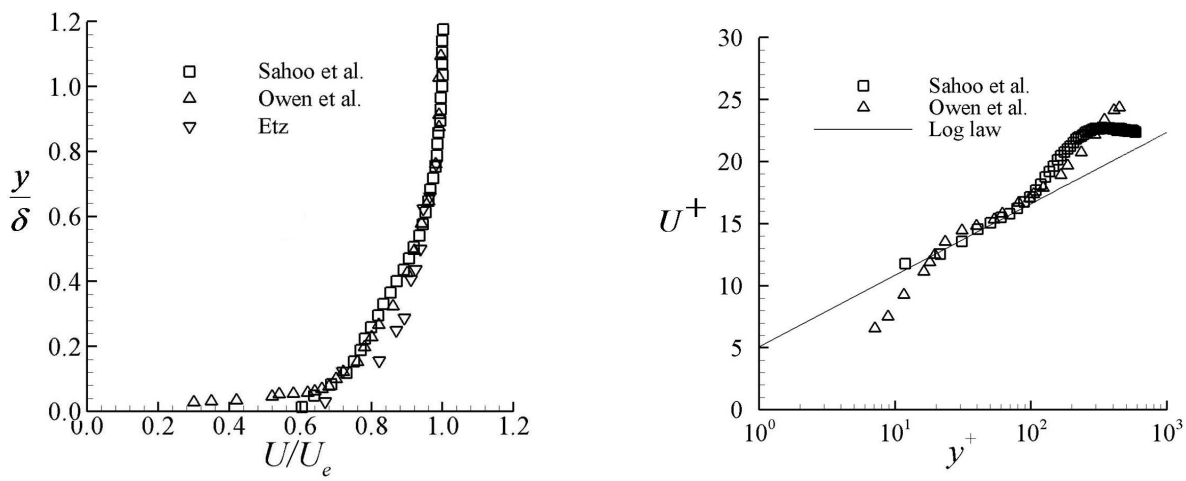

Figure 5. Left: Comparison of the Pitot probe data of Owen et al.[21] at Mach 6.7 with PIV data obtained at Mach 7.2. Right: Law-of-the-wall correlation using Van Driest transformation. The log-law is given by Equation 2.

The streamwise velocity profiles for the smooth, mesh, and square bar roughness cases are shown in Figure 6. When transformed according to Van Driest, the profiles demonstrate the expected effects of surface roughness, whereby the profiles are shifted below the standard log-law by $\Delta(U/u_\tau) = 13.8$ and 13 for the mesh and square bar roughness, respectively. This is somewhat surprising, given the large roughness heights employed here ($y/\delta = 0.16$ and 0.14 for the mesh and square bar surfaces, respectively).

The profiles in outer layer scaling are shown in Figure 7. According to Coles,

$$\frac{U_e - U}{u_\tau} = -\frac{1}{\kappa}\ln\left(\frac{y}{\delta}\right) + \frac{2\Pi}{\kappa}\left(1 - \sin^2\left(\frac{\pi}{2}\frac{y}{\delta}\right)\right) \tag{4}$$

Here we assume that this correlation still applies when the velocity transformed according to van Driest, and when δ is measured from the virtual origin of the velocity profile.

The smooth wall profile agrees reasonably well with the Coles wake function, although there is a small discrepancy in the middle of the layer that was also seen in Figure 5. The rough-wall profiles display a somewhat smaller wake function, but they agree well with each other.

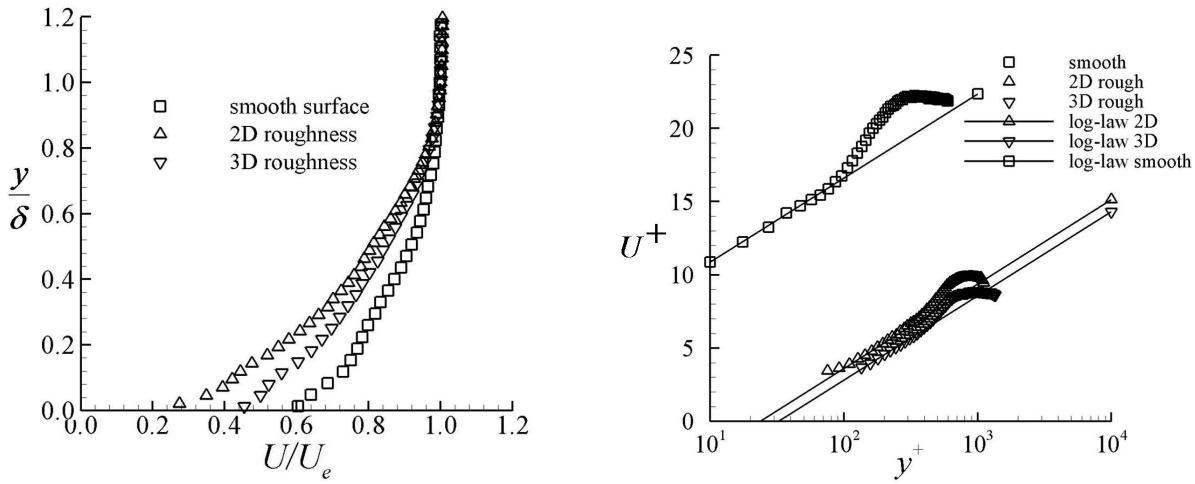

Figure 6. Left: Mean velocity profile of smooth wall (transformed according to Van Driest) compared to diamond mesh (3D) and square bar (2D) roughness. Right: Law-of-the-wall correlation using Van Driest transformation. The log-law is given by Equation 2.

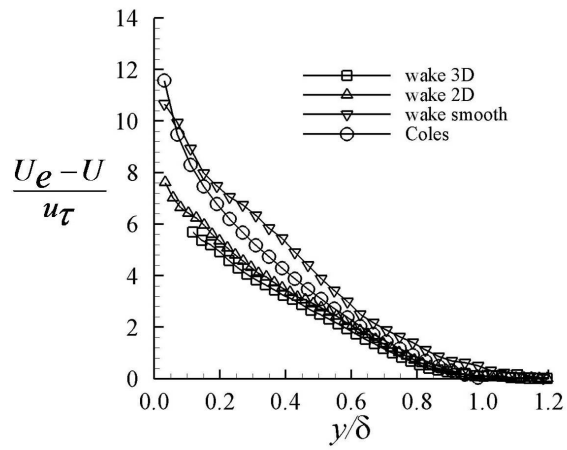

Figure 7. Wake profile of smooth wall (transformed according to Van Driest) compared to diamond mesh (3D) and square bar (2D) roughness, in outer scaling. The curve labeled Coles is according to Equation 4.

V. Turbulence Results

The streamwise velocity fluctuations for the smooth plate are shown in Figure 8. They agree well with Owen *et al.*[21] results for $y/\delta > 0.16$, using either classic or Morkovin scaling. However, the hypersonic results fall well below similar incompressible data. Given that the two data sets were obtained by very different means (one by hot wires and transformed using the SRA, the other by direct measurement using PIV), we appear to be seeing significant compressibility effects at these Mach numbers.

The streamwise and wall-normal velocity fluctuations for smooth plate are shown in Figure 9. The wall-normal component in the region $0 \leq \delta < 0.4$ shows a strong damping, which may be indicative of low Reynolds number effects, although measurement errors may also be important in this region.

The turbulent intensities for the mesh and square bar roughness are shown in Figure 10 using classic scaling. Surprisingly, the turbulence intensities are strongly damped by the effects of roughness, and this trend is even more apparent when the results are scaled according to Morkovin, as seen in Figure 11. Note, however, that the results represent streamwise averages over the field of view, which is 22 mm long, equivalent to about 2.7λ for both rough surfaces. As Ekoto *et al.*[5] noted, there exist large flow variations within a distance λ for a mesh-type roughness, associated with the formation of local shocks and expansion fans. At

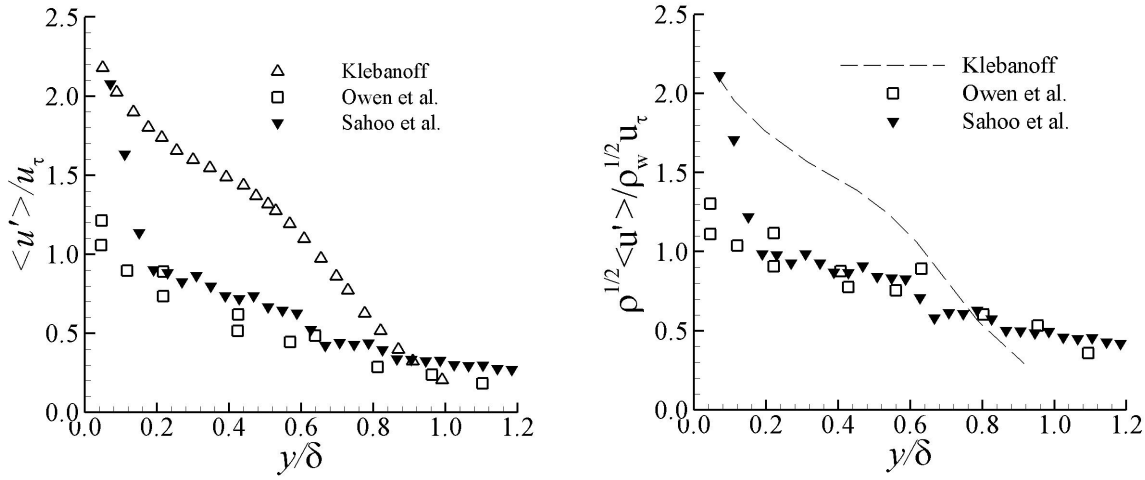

Figure 8. Left: Distribution of streamwise velocity fluctuations across the boundary layer. Right: Morkovin's co-ordinate stretching density factor applied to the streamwise velocity fluctuations.

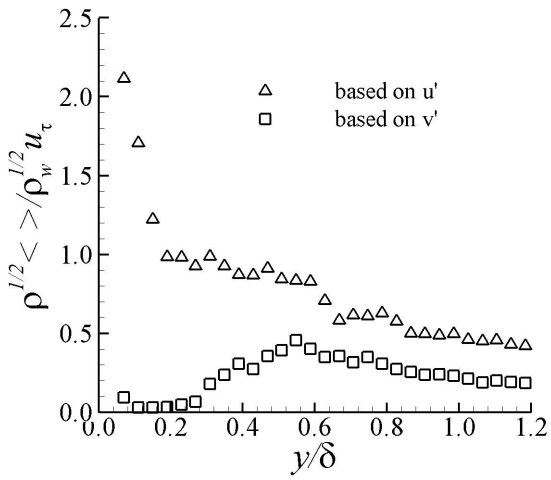

Figure 9. Morkovin's co-ordinate stretching density factor applied to streamwise and wall-normal velocity fluctuations for smooth plate.

their Mach number (2.86), their streamwise-averaged turbulence intensities for the square block roughness did not show any significant compressibility effects (associated with small axial flow variations), but their diamond mesh roughness displayed such large streamwise variations in $\overline{u'^2}$ and $\overline{v'^2}$ (about $\pm 30\%$) that they did not present streamwise averaged turbulence results for this roughness. The alternating velocity gradients produced production fields that produced and subsequently damped the turbulent stresses. As they pointed out, strong localized disturbances may provide a mechanism for flow manipulation.

It should also be noted that the hot wire results of Berg[1] at Mach 6 showed that, in the equilibrium region well downstream of the smooth to rough transition, his largest square bar roughness (where $k^+ = 33.8$, assumed to be fully rough) amplified the rms mass-flux fluctuations by a factor of about two near $y/\delta = 0.5$, but actually showed some damping for $y/\delta < 0.1$. It is clear that we need further information on the velocity and density fields in order to understand this behavior more fully.

VI. Conclusions

PIV experiments were conducted to study the effects of surface roughness on a hypersonic turbulent boundary layer at DNS-accessible Reynolds numbers. Diamond mesh and square roughness geometries were

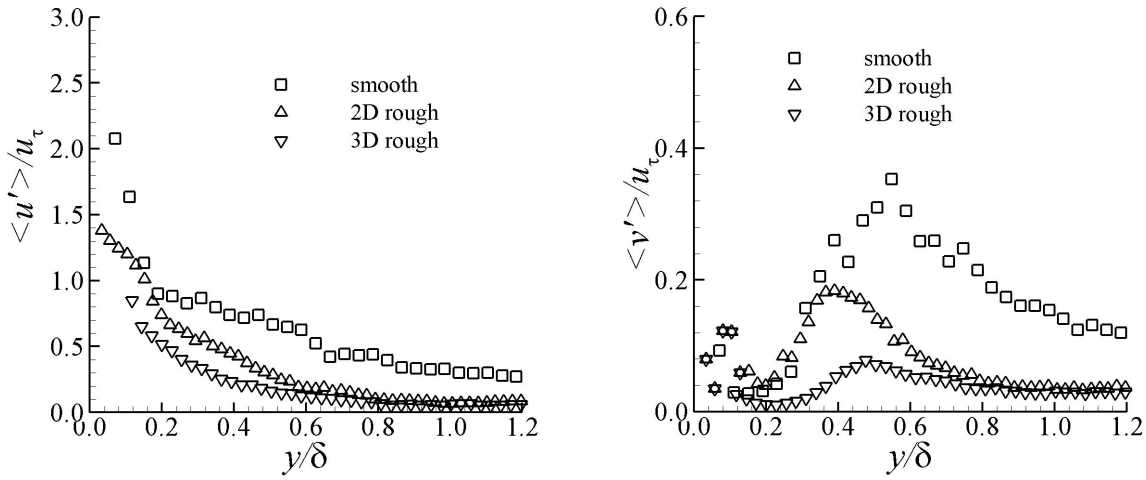

Figure 10. Velocity fluctuations for smooth, 3d roughness, and 2d roughness case. Left: Axial velocity fluctuations Right: transverse velocity fluctuations.

compared to an aerodynamically smooth wall.

For the case of a smooth wall, the mean velocity profile (transformed according to Van Driest) agreed well with previous results, regardless of the Mach number. The turbulence intensities were significantly smaller than seen in incompressible flows, but they agreed well with previous data at the same Mach number taken using very different techniques, which suggested a strong effect of compressibility on the velocity fluctuations.

The mean velocity profiles for both rough walls showed the expected response to roughness, which is the shift of the velocity profile below the standard log-law, without a change in shape. The magnitude of the wake parameter decreased, but this may be due to transitional effects.

The streamwise and wall-normal velocity fluctuations for the rough walls were strongly damped. This result is unexpected, but may be associated with the local flow distortions caused by the roughness, as suggested by the earlier study of Ekoto *et al.*[5] at Mach 2.86.

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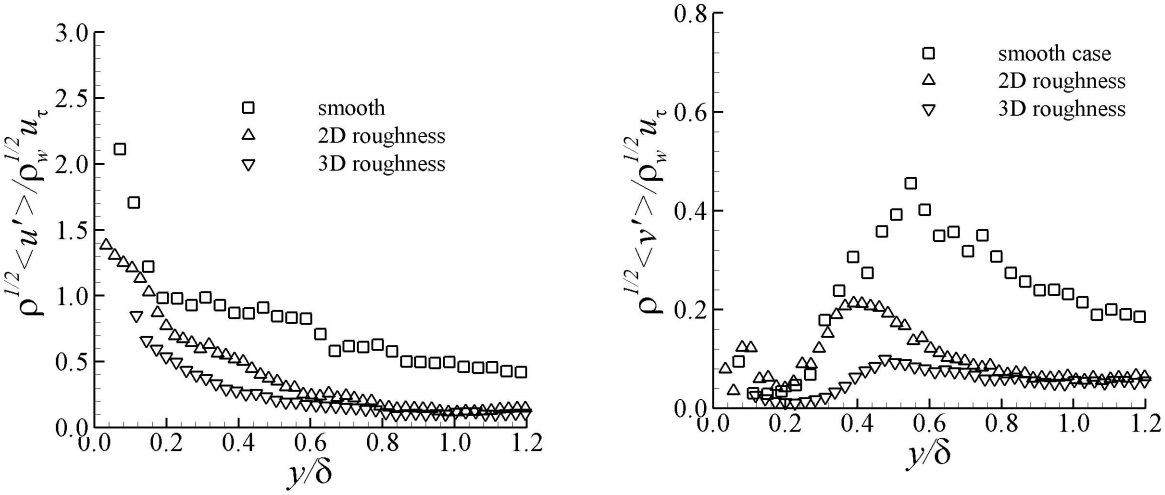

Figure 11. Morkovin's co-ordinate stretching density factor applied to velocity fluctuations for smooth, 3d roughness, and 2d roughness case. Left: Axial velocity fluctuations Right: transverse velocity fluctuations.

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